NLP Patient Description to Acuity

Presentation Q&A Guide — Detailed Answers

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Contents

1 Introduction

This guide contains detailed answers to every question asked during preparation for the **NLP Patient-to-Acuity** presentation (`slides_new/nlp_patient_to_acuity.pdf`, 28 slides, 16:9 Beamer).

Running example throughout:

> *"I have a headache and feel feverish. My body aches and I feel weak."*

2 Clinical Context (Pages 3–5)

2.1 Page 4 — Fusion safety rule

QUESTION

What does it mean that fusion assigns a less urgent colour than a strong signal requires?

ANSWER

Curatio uses **two parallel pathways**: TEWS (vitals) and NLP (patient words). Fusion combines them into a final SATS colour.

The safety rule means: if BioBERT assigns a **strong, confident** urgency signal from language (e.g. crushing chest pain, severe distress), fusion must **not downgrade** to a less urgent colour (e.g. Green) simply because vitals look normal.

Fusion is a **priority checklist**, not an average of scores. Strong language can pull the final colour *upward*; weak vitals alone cannot pull a strong language signal below the safety floor. This prevents **under-triage** — the most dangerous error in emergency care.

HOW TO SAY IT LIVE

"We never under-triage when the words scream emergency. Vitals supplement the text — they don't override a strong language signal."

2.2 Page 5 — Bag of words and supervised learning

QUESTION

What does bag-of-words mean? How should I explain page 5 during my presentation?

ANSWER

Bag-of-words (BOW): Throw all words into a bag, count frequencies, feed counts to a classifier. Word order and negation are lost. "Not feverish, no headache" and "headache and feverish" can look similar because they share vocabulary.

Supervised deep learning (our approach):

$$X \xrightarrow{\tau} \text{tokens} \xrightarrow{E} \text{embeddings} \xrightarrow{\text{12 layers}} h_{[\text{CLS}]} \xrightarrow{\text{softmax}} \text{acuity}$$

Labels come from nurses (Triagegeist, ~80,000 chief complaints). The model learns context bidirectionally — "feverish" modifies "headache" through self-attention.

HOW TO SAY IT LIVE

"Old NLP counted words in a bag and lost order. Our BioBERT reads the full sentence, learns from 80,000 nurse-labeled complaints, and predicts acuity with context."

3 NLP Pipeline (Pages 6–19)

3.1 Page 6 — Tokenization

QUESTION

What does τ mean? What is t representing? Why is $M \leq 128$? Why ∼30,000 vocabulary? Why row index? How is the ID assigned? What is embedding?

ANSWER

- $\boldsymbol{\tau}$ — the **tokenizer** function: splits raw text X into sub-word tokens $(t_1, t_2, \ldots)$.
- t_i — the i-th token (word-piece) after tokenisation.
- $M \leq 128$ — hard cap on sequence length; longer complaints are truncated. Keeps inference fast (attention cost grows with length^2).
- ∼30,000 vocab (28,996 for BioBERT) — fixed dictionary from pre-training.
- **Row index** — each token ID is the row number in the embedding table E: ID 1045 ⇒ row 1045 ⇒ 768-d vector.
- **ID assignment** — tokenizer matches text to vocabulary; known words get fixed IDs; rare words split into sub-pieces (e.g. "##ish").
- **Embedding** — lookup table E: integer ID → 768-number vector (page 9).

3.2 Pages 7–8 — UNK, brackets, [CLS], $h_{\text{[CLS]}}$

QUESTION

What do the square brackets in [CLS] mean? What happens when UNK is assigned? Why is [CLS] = 101? What does the h in $h_{\text{[CLS]}}$ mean?

ANSWER

- **Square brackets** [] — mark **special tokens**, not ordinary words. [CLS], [SEP], [MASK] are control tokens.
- <redacted_UNK> (ID 100) — assigned when the tokenizer cannot match a piece. The model still runs but may lose fine detail for that word.
- **Why [CLS] = 101** — fixed ID from BERT's original vocabulary design; arbitrary but consistent across all inputs.
- **h in $h_{\text{[CLS]}}$** — h = **hidden state**: the 768-d vector for the [CLS] token after all 12 layers. Row 0 of $H^{(12)}$.

[CLS] is prepended to every sentence; its final state is the complaint summary used for classification.

3.3 Page 9 — Embedding lookup

QUESTION

Explain slide 9. When I pick "headache", it has 768 numbers — where are they coming from?

ANSWER

Slide 9 shows **embedding lookup**: token ID $\xrightarrow{E}$ 768-dimensional vector.
The 768 numbers are **learned weights**, not extracted from PubMed text. Each vocabulary row holds 768 floats updated during:

1. MLM pre-training on PubMed/PMC (∼18B words)
2. Fine-tuning on Triagegeist (∼80k labeled complaints)

D1, D2, . . . , D768 are dimension labels — not named clinical features. The model discovers which combinations encode medical meaning.

3.4 Cross-cutting — PubMed and BioBERT paper

QUESTION

Where is the PubMed training data? How do the 768 features look in the data? What do D1, D2 numbers mean?

ANSWER

PubMed is not one downloadable table — it is millions of biomedical abstracts at `https://pubmed.ncbi.nlm.nih.gov/`. Full-text articles: `https://www.ncbi.nlm.nih.gov/pmc/`.
BioBERT paper: `https://academic.oup.com/bioinformatics/article/36/4/1234/5566506`
The 768 features do **not appear as columns in PubMed text**. They are **internal weight matrix rows** learned by gradient descent. You cannot open a CSV and see D1–D768 for "headache" — they exist only in the model checkpoint (`model.safetensors`).
Each D_j is one coordinate in learned semantic space. After training, similar clinical terms cluster near each other.

3.5 Page 11 — Contextual mapping

QUESTION

Explain page 11.

ANSWER

Before entering the transformer, each token vector is the **sum of three embeddings**:

$$H^{(0)}[i] = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(i)$$

- **Token embedding** — what word it is ("headache").
- **Position embedding** — where it sits (1st, 2nd, . . .).
- **Segment embedding** — which sentence segment (usually 0 for one complaint).

The diagram shows three vectors adding point-wise. Same word at different positions gets different input vectors.

3.6 Page 13 — Input matrix and why 768

QUESTION

Explain page 13. What is 768 and why that number?

ANSWER

$H^{(0)}$ is a matrix: **one row per token, 768 columns**. Shape [seq_len $\times$ 768]. This is the input to encoder layer 1.
Why 768? BERT-Base architecture standard:

- `hidden_size` = 768
- 12 layers, 12 attention heads
- 64 dims per head $\times$ 12 heads = 768

BioBERT inherits this from BERT — not tuned specifically for triage. It balances capacity vs. speed.

3.7 Page 15 — Encoder diagram, arrows, FFN

QUESTION

Explain page 15. Why do arrows point upward? What is a feed-forward network?

ANSWER

Upward arrows: standard diagram convention — input tokens at the **bottom**, output after 12 layers at the **top**. Data flows upward through the stack.
Feed-Forward Network (FFN): Two linear layers with ReLU:

$$\mathrm{FFN}(x) = W_2\,\mathrm{ReLU}(W_1 x + b_1) + b_2$$

Expands 768 $\to$ 3072 $\to$ 768. Adds non-linear processing *per token* after attention mixes context. Like two $y = Wx + b$ lines with a ReLU bend between them.

3.8 Page 16 — Z^ℓ and H^ℓ

QUESTION

Explain slide 16. Explain the FFN formula using the equation of a line. Use simple mathematics for Z^ℓ and H^ℓ.

ANSWER

Inside each encoder layer ℓ:

$$Z^{(\ell)} = \text{LayerNorm}\Big(H^{(\ell-1)} + \text{MultiHeadAttention}(H^{(\ell-1)})\Big)$$
$$H^{(\ell)} = \text{LayerNorm}\Big(Z^{(\ell)} + \text{FFN}(Z^{(\ell)})\Big)$$

FFN as equation of a line: First line $y_1 = W_1 x + b_1$, ReLU gate (zero out negatives), second line $y_2 = W_2 y_1 + b_2$.
Simple numeric walkthrough: Suppose $H^{(\ell-1)}$[headache] $= 2.0$ and attention adds 0.5 from "feverish":

- Pre-norm sum: $2.0 + 0.5 = 2.5$
- LayerNorm scales to e.g. $2.3 \Rightarrow$ part of $Z^{(\ell)}$
- FFN transforms $2.3 \to 2.8$; residual: $2.3 + 2.8 = 5.1$; LayerNorm $\Rightarrow H^{(\ell)}$[headache]

Residual connections keep gradients stable during training.

3.9 Pages 18–19 — Self-attention and W matrices

QUESTION

Explain slides 18 and 19. What does the W there mean?

ANSWER

Slide 18 (diagram): Four steps — (1) split embeddings into Q, K, V; (2) score with $QK^\top$; (3) softmax to probabilities; (4) blend Value vectors. For "headache": feverish 0.41, headache 0.32, weak 0.18, filler ≈ 0.02.
Slide 19 (formula):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

W_Q, W_K, W_V: Learned weight matrices (like $y = Wx+b$). Same token matrix H multiplied by three different W gives three roles:

- W_Q — "what is this token looking for?"
- W_K — "what does each token advertise?"
- W_V — "what content to pass forward?"

Each is $\sim 768 \times 64$ per head; 12 heads in parallel. Updated via backpropagation during training.

4 BERT Pre-Training (Pages 20–22)

4.1 Page 20 — Masked Language Modeling

QUESTION

Explain slide 20.

ANSWER

Before triage fine-tuning, BioBERT learns medical English via **Masked Language Modeling (MLM)** on PubMed/PMC:

- 15% of tokens randomly masked with `[MASK]`
- Model predicts missing words from **bidirectional** context
- Teaches vocabulary ("feverish", "tachycardia") **without** acuity labels

BioBERT = BERT architecture + biomedical pre-training (∼18B words). Fine-tuning on Triagegeist comes later.

4.2 Page 21 — MLM loss

QUESTION

Explain slide 21.

ANSWER

$$\mathcal{L}_{\mathrm{LM}} = -\sum_{i \in \mathcal{M}} \log P(t_i \mid H^{(L)}; \theta)$$

- $\mathcal{M}$ — masked positions; t_i — true word; $H^{(L)}$ — context from 12 layers; θ — all weights.
- $P = 0.92 \Rightarrow -\log(0.92) \approx 0.08$ (small penalty)
- $P = 0.01 \Rightarrow -\log(0.01) \approx 4.6$ (large penalty)

Same $-\log P$ pattern as triage cross-entropy — only the label changes.

5 Learning Math (Pages 23–27)

5.1 Page 23 — Attention softmax

QUESTION

Explain page 23.

ANSWER

$$\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^{m} \exp(e_k)}$$

Converts raw attention scores e_j into probabilities α_j summing to 1. Used **inside** self-attention (over tokens), not for classification.

exp forces positive values and amplifies large scores. "I", "a" $\to \alpha \approx 0.01$; "headache", "feverish" $\to \alpha \approx 0.35+$.

5.2 Page 24 — Classification softmax

QUESTION

Explain page 24.

ANSWER

$$\hat{\mathbf{y}} = \mathrm{softmax}(W h_{[\mathrm{CLS}]} + \mathbf{b}), \quad \hat{y}_c = P(\text{acuity} = c \mid X)$$

- $h_{[\mathrm{CLS}]}$ — 768-d summary from layer 12
- $W \in \mathbb{R}^{5\times 768}$ — learned classification weights
- $\hat{y}_c$ — probability of acuity level c; $\sum_c \hat{y}_c = 1$

Running example: Level 3 (Moderate) = 72% ⇒ Yellow on SATS. Class indices 0–4 map to acuity levels 1–5 in the deployed Curatio model.

5.3 Page 25 — Forward pass and cross-entropy

QUESTION

Explain page 25.

ANSWER

1. **Forward pass:** $z = Wx + b$ where $x = h_{[\mathrm{CLS}]}$, then $\hat{y} = \mathrm{softmax}(z)$.
2. **Cross-entropy loss:**

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i), \quad C = 5$$

With one-hot nurse label, collapses to $-\log(\hat{y}_{\text{correct}})$.
Example: true = Yellow, model predicts Green at 90% (Yellow only 5%) ⇒ $\mathcal{L} = -\log(0.05) \approx 3.0$.

5.4 Page 26 — Backpropagation

QUESTION

Explain page 26.

ANSWER

$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial W}$$

Chain rule traces error backward: loss → softmax → logits $z = Wh + b$ → all 12 layers → embeddings.
Each $\partial\mathcal{L}/\partial W$ pinpoints how much weight W contributed to the mistake. Example: model says Green 90% when nurse labeled Yellow — backprop finds weights in layers 8–12 that over-weighted non-urgent patterns.

5.5 Page 27 — Gradient descent

QUESTION

Explain page 27.

ANSWER

$$W_{\text{new}} = W_{\text{old}} - \alpha \frac{\partial \mathcal{L}}{\partial W}$$

- $\alpha = 2 \times 10^{-5}$ — learning rate (`triage_new.ipynb`)
- Minus sign — move *opposite* to gradient (downhill on loss)
- $\mathcal{L}_{\text{triage}} = -\sum_{i=1}^{N} \log P(y_i \mid X_i; \theta)$, $N \approx 80{,}000$
- Best `eval_loss` = 0.001812 after 3 epochs, 13,500 steps

6 Appendix

6.1 Data and model references

- **Triagegeist (fine-tuning):** `https://www.kaggle.com/competitions/triagegeist/data`
- **PubMed abstracts:** `https://pubmed.ncbi.nlm.nih.gov/`
- **PMC full text:** `https://www.ncbi.nlm.nih.gov/pmc/`
- **BioBERT paper:** `https://academic.oup.com/bioinformatics/article/36/4/1234/5566506`

6.2 BioBERT-Base configuration

Parameter	**Value**
hidden_size	768
num_hidden_layers	12
num_attention_heads	12
vocab_size	28,996
max_length	128
num_labels (fine-tuned)	5
learning_rate	2×10^{-5}
training epochs	3

6.3 Acuity to SATS mapping (Curatio)

Class index	**Acuity level**	**SATS colour**
0	1	Red
1	2	Orange
2	3	Yellow
3	4	Green
4	5	Green